

Week 2 Report

Progress this week:

- Reviewed the current Augenblick architecture and traced image preparation, pose estimation, reconstruction, mesh export, and evaluation
- Confirmed that all reconstruction methods consume a common COLMAP scene containing images, optional masks, and a sparse camera model
- Reviewed the Linux, CUDA, PACE, and HiPerGator environment requirements
- Audited the current color pipeline: images are not ColorChecker-calibrated, and existing exposure compensation and appearance metrics do not establish true color accuracy
- Confirmed that coded-marker detection and conversion to millimetres are not currently implemented
- Reviewed color-calibration literature and identified ColorChecker-based correction with CIEDE2000 evaluation as the appropriate direction
- Reviewed the Imageomics package: it uses COCO card masks and RGB histogram matching to improve internal consistency, but does not use official ColorChecker values
- Proposed a small first deliverable: calibrate one capture session, preserve the original images, and report color error before and after correction

Next Week

- Arrange a working session with Om and divide the investigation clearly so that we do not duplicate effort.
- Compare the existing Imageomics color-calibration tool's inputs and assumptions with the museum images and the current Augenblick workflow.
- Confirm the ColorChecker model and access to representative photographs, chart regions, segmentation masks, and the required execution environment.
- Run one small representative calibration test if the required data and environment are available.
- Document the adaptations, missing inputs, blockers, and likely integration points, and contact Arthur midweek if a decision is needed.
- Bring a clear recommendation to the next meeting: adapt the existing tool, or explain why it is unsuitable and propose the next alternative.

Abstracts

Preserving Colour Fidelity in Photogrammetry - An Empirically Grounded Study and Workflow for Cultural Heritage Preservation

- This study examines how image acquisition, RAW development, color-space selection, ColorChecker calibration, and photogrammetric processing affect the color fidelity of cultural-heritage 3D models. It recommends controlling image transformations, calibrating photographs with known chart values, and evaluating color differences with CIEDE2000. The paper supports treating color accuracy as a separate measurement problem rather than relying only on visual similarity or reconstruction metrics.

Imageomics Color Calibration Pipeline

- The Imageomics package calibrates a dataset by choosing one photograph as a reference, extracting the color card from each annotated image, and matching the target card's red, green, and blue histograms to the reference card. The resulting lookup tables are applied to each full photograph and saved as PNG files. This can reduce variation between images, but it provides relative normalization rather than absolute ColorChecker calibration because it does not use official patch values or report CIEDE2000 error.